ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
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Department of Computer Science and Engineering (CSE)

Project Report

SPL-I SWE 4304 : Software Project Lab I

Project Title

AirCLI

Group 3

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Project Overview

AirCLI is a command-line–based airport management system that simulates the complete passenger journey—from account creation and flight booking to baggage handling and airport security checks.

The system focuses solely on **customer-facing airport operations**, including:

- User authentication and payments
- Flight booking and ticketing
- Baggage packing and security inspection
- Airport staff workflows (ticket checking, baggage handling, security escalation) only relevant to the customer

Motivation

The motivation behind **AirCLI** is to design and implement a system that realistically models customer-facing airport operations using fundamental programming principles.

A key challenge in airport systems is handling baggage security without relying on intrusive or unrealistic assumptions, such as passengers voluntarily declaring illegal items. This project addresses that challenge by introducing a **two-level baggage packing model**, where passengers can pack items normally while concealed items remain inaccessible until a security escalation occurs. This design enables the simulation of realistic inspection and enforcement processes without randomness or external hardware.

Another motivating factor is the desire to explore **role-based authority and decision-making** within a software system. By clearly separating responsibilities among passengers, baggage handlers, senior security officers, and administrators, the project demonstrates how access control and state transitions can be enforced programmatically to prevent unauthorized actions.

Overall, **AirCLI** serves as a practical learning opportunity in building a modular software system that mirrors real-world airport workflows. The project highlights how complex operational logic—such as baggage inspection, ticket validation, and security escalation—can be implemented using basic programming constructs, making it both an educational exercise and a realistic system simulation.

Flowchart

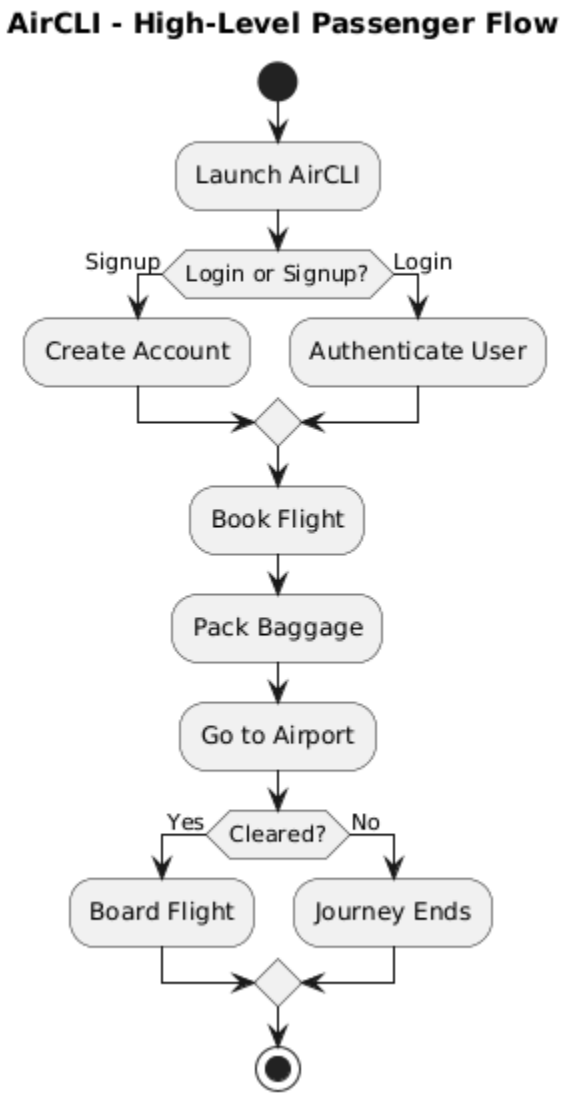

Fig: Passenger Flow

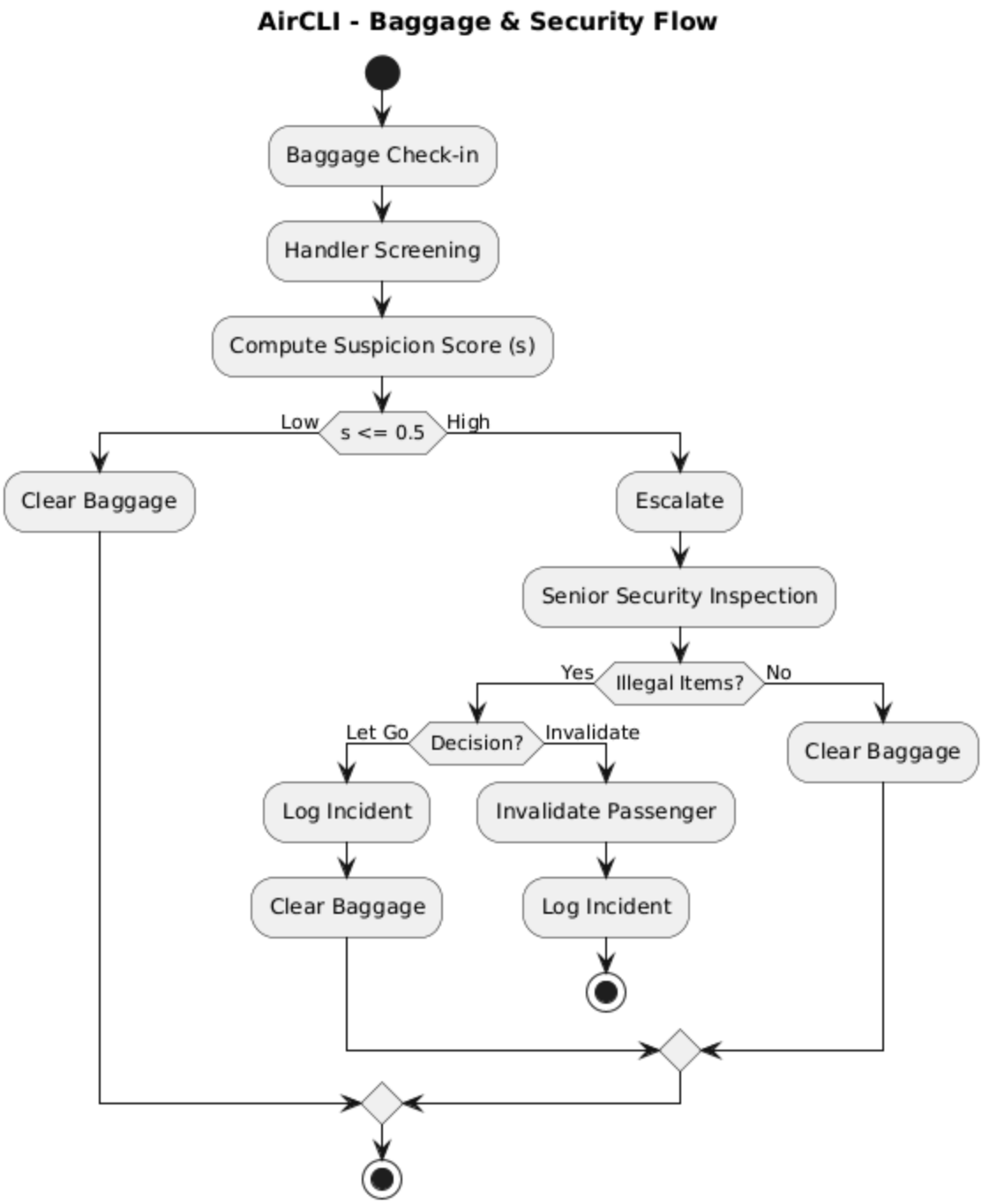

Fig: Baggage and Security Flow

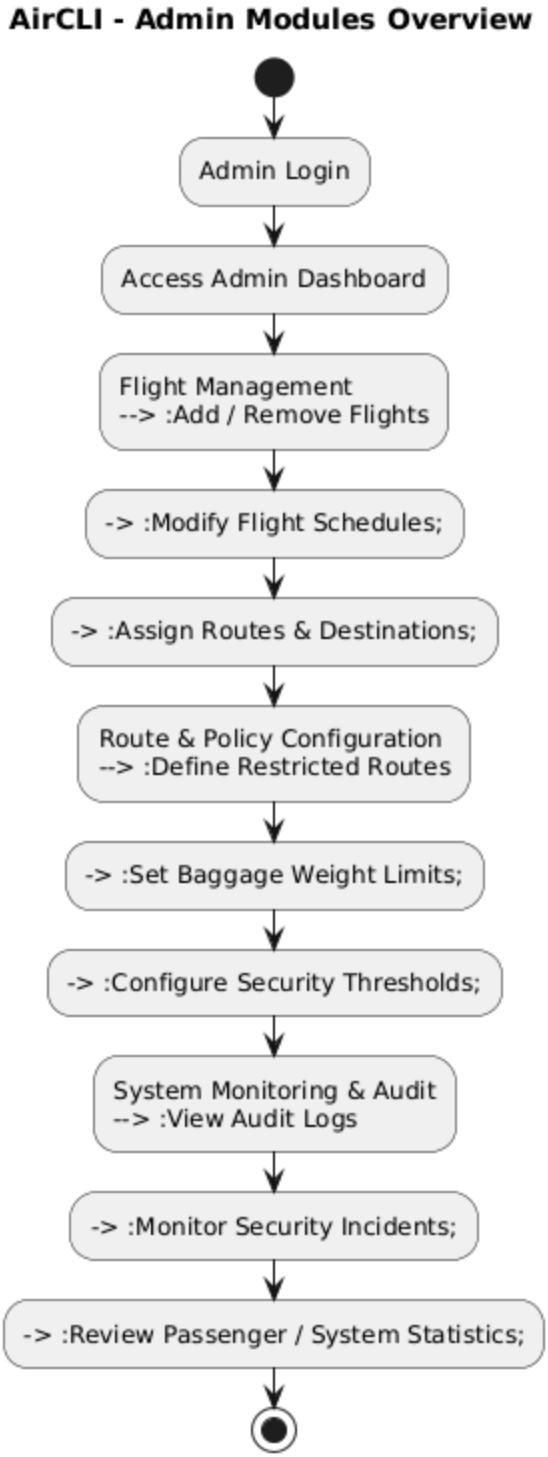

Fig: Admin Module Flow

Key Features

1. User Roles

Role	Description
Passenger	Books flights, packs baggage, completes airport procedures
Baggage Handler	Performs preliminary baggage checks and risk assessment
Senior Security Officer	Conducts deep inspection and enforcement actions
Admin	Manages flights, routes, and system configuration

2. Core Modules

2.1. User-Side Modules (Passenger-facing)

These modules represent all functionalities directly accessible to passengers through the CLI.

2.1.1. User Account Module (airCLI)

The User Account Module manages passenger identity within the system. It supports user registration and authentication, maintains user credentials, and associates each passenger with their booking history, baggage records, and linked banking information. This module serves as the entry point for all passenger-side interactions.

- User registration (signup)
- Authentication (login)
- Profile Viewing
- Role Assignment (passenger)

2.1.2. Banking and Payment Module (Bank CLI)

The Banking and Payment Module simulates financial transactions required for flight bookings. It validates payment credentials, verifies account balances, processes debits and credits, and records transaction history. The module ensures that bookings proceed only after successful and traceable payment confirmation.

- Simulated Bank Account Linkage
- PIN Validation
- Balance Management
- Payment Processing for Flight Bookings
- Transaction Logging

2.1.3. Flight Search & Booking Module (airCLI)

This module enables passengers to discover and reserve flights based on travel criteria such as origin, destination, and date. It handles flight selection, coordinates with the payment system, and initiates ticket creation upon successful booking. The module forms the core of the passenger travel planning process.

- Search flights by: Origin, Destination, Date
- Flight Selection
- Payment Initiation
- Ticket Issuance after Successful Payment

2.1.4. Ticket Management Module (airCLI)

The Ticket Management Module is responsible for generating, storing, and maintaining ticket records. It tracks ticket states such as validity, expiration, early arrival, and invalidation due to security actions. This module ensures that each ticket accurately reflects the passenger's authorization to travel.

- Ticket Generation
- Ticket Storage
- Ticket Status Tracking (Valid, Expired, Too Early, Invalid)
- Ticket-to-User Mapping

2.1.5. Baggage Packing Module (Home CLI)

The Baggage Packing Module simulates the process of preparing luggage prior to airport arrival. It allows passengers to organize baggage contents using a two-level structure, separating declared items from private items. The module calculates baggage weight and prepares

baggage data for subsequent airport-side inspection.

- Baggage creation
- Two-level item loading: Declared, Hidden
- Weight Calculation
- Final Baggage Submission

2.1.6. Passenger Activity Module

This module provides optional informational and navigational features for passengers. It allows users to view upcoming flights, review past bookings, monitor baggage status, and transition into the airport phase of the system. The module enhances usability without influencing core security decisions.

- View Upcoming Flights
- View Travel History
- Track Baggage Status
- Enter Airport Mode

2.2. Security-Side Modules (Airport Authority)

2.2.1. Ticket Checking Module

The Ticket Checking Module operates at the airport entry point to verify passenger authorization. It validates ticket authenticity and checks time-based constraints such as early arrival or expiration. Based on the evaluation, the module either permits entry and assigns a gate or denies access.

- Ticket Authentication
- Time-based Validation
- Platform/ Gate Assignment
- Entry Authorization or Rejection

2.2.2. Baggage Screening Module (Handler)

The Baggage Screening Module conducts preliminary, non-invasive inspection of passenger luggage. It evaluates factors such as weight irregularities, route-item inconsistencies, and passenger security history to compute a suspicion score. This module determines whether baggage is cleared or escalated for deeper inspection.

- Non-invasive baggage inspection

- Weight Mismatch Detection
- Route-item consistency Evaluation
- Passenger History Lookup
- Suspicion Score Calculation
- Escalation Decision

2.2.3. Senior Security Module

The Senior Security Module represents the highest level of inspection authority. It is activated only for escalated baggage cases and has exclusive access to concealed baggage contents. The module performs deep inspections and determines the final outcome, either clearing the passenger or invalidating their journey, while logging all decisions for audit purposes.

- Role-based Authentication
- Access to Hidden Baggage Layer
- Deep Baggage Inspection
- Illegal Item Identification
- Final Security Decision (Clear Passenger, Invalidate Passenger)
- Incident Logging

2.3. Admin-Side Modules (System Control)

2.3.1. Flight Management Module

This module enables administrators to manage flight-related data within the system. It supports the creation, modification, and removal of flights, as well as schedule and destination management. The module ensures that accurate and up-to-date flight information is available for booking and validation.

- Add/ Remove Flights
- Modify Flight Schedules
- Assign Route/ Destination

2.3.2. Route & Policy Config Module

The Route and Policy Configuration Module allows administrators to define operational rules such as restricted routes, baggage weight limits, and security thresholds. These configurations influence screening decisions and ensure consistent enforcement of airport policies across the system.

- Define Restricted Routes
- Configure Baggage Limits
- Set security thresholds

2.3.3. System Monitoring Module

This module provides administrative visibility into system operations. It allows administrators to review logs, monitor security incidents, and observe overall system activity. The module supports oversight and accountability without modifying operational data.

- View Audit Logs
- Monitor Security Incidents
- Review Passenger Statistics

2.4. General Modules (Non-Role-Specific)

2.4.1. Data Storage Module

The Data Storage Module manages persistent storage using structured text files. It handles reading, writing, and updating system data including users, flights, tickets, baggage records, and logs. The module ensures data consistency without reliance on external databases.

- Text-file-based persistence
- Structured File Formats
- Data loading and saving
- Backup Handling

2.4.2. Access Control Module

The Access Control Module enforces role-based permissions across the system. It ensures that users, airport staff, and administrators can only access functionality appropriate to their assigned roles. This module prevents unauthorized actions and protects sensitive system data.

- Role Validation
- Permission Enforcement
- Restricted data access

2.4.3. Audit and Logging Module

This module records all critical system actions in append-only logs. It tracks financial transactions, baggage inspections, and security decisions

to ensure traceability and accountability. Logged data supports review, debugging, and compliance verification.

- Payment Logs
- Baggage Inspection Logs
- Security Incident Logs

2.4.4. Utility and Validation Module

The Utility and Validation Module provides shared support functions across the system. It handles input validation, date and time checks, identifier generation, and consistency verification. This module improves reliability and reduces duplication of common logic.

- Input Validation
- Date & Time Validation
- ID Generation
- Consistency Checks

Tools & Technology

1. IDE: Visual Studio Code
2. Compiler: GNU G++
3. Version Control: Git
4. Diagrams (UML or Flowchart): PlantUML

Proposed Timeline

Phase	Subtask	W 1	W 2	W 3	W 4	W 5	W 6	W 7	W 8	W 9	W 10	W 11	W 12	W 13	W 14
Planning Design	Requirement Analysis														
	Project Presentation														
	System Design & Architecture														
	UML & Flow Diagrams														
Core User Modules	User Account Module														
	Banking & Payment Module														
	Flight Search & Booking														
	Ticket Management														
Baggag e & Security	Baggage Packing Module														
	Baggage Screening Module														
	Senior Security Module														

Phase	Subtask	W 1	W 2	W 3	W 4	W 5	W 6	W 7	W 8	W 9	W 10	W 11	W 12	W 13	W 14
Airport Operations	Ticket Checking Module														
	Passenger Activity Module														
Admin /System Module	Flight Management Module														
	Route & Policy Configuration														
	Access Control & Logging														
Testing & Documentation	Integration Testing														
	Documentation														

Actions Taken Based on Feedback

1. It was mentioned that our feature explanation required some work, so we took time to ensure that all the issues mentioned are properly addressed.

2. Our Tools & Technology part was incomplete, as it only had the frameworks we were using. We improved on it by including the IDE, Version Control, and Diagram tool that we are using for a robust overview.

3. The Gantt chart was missing from our initial presentation, which has now been included in the final report.